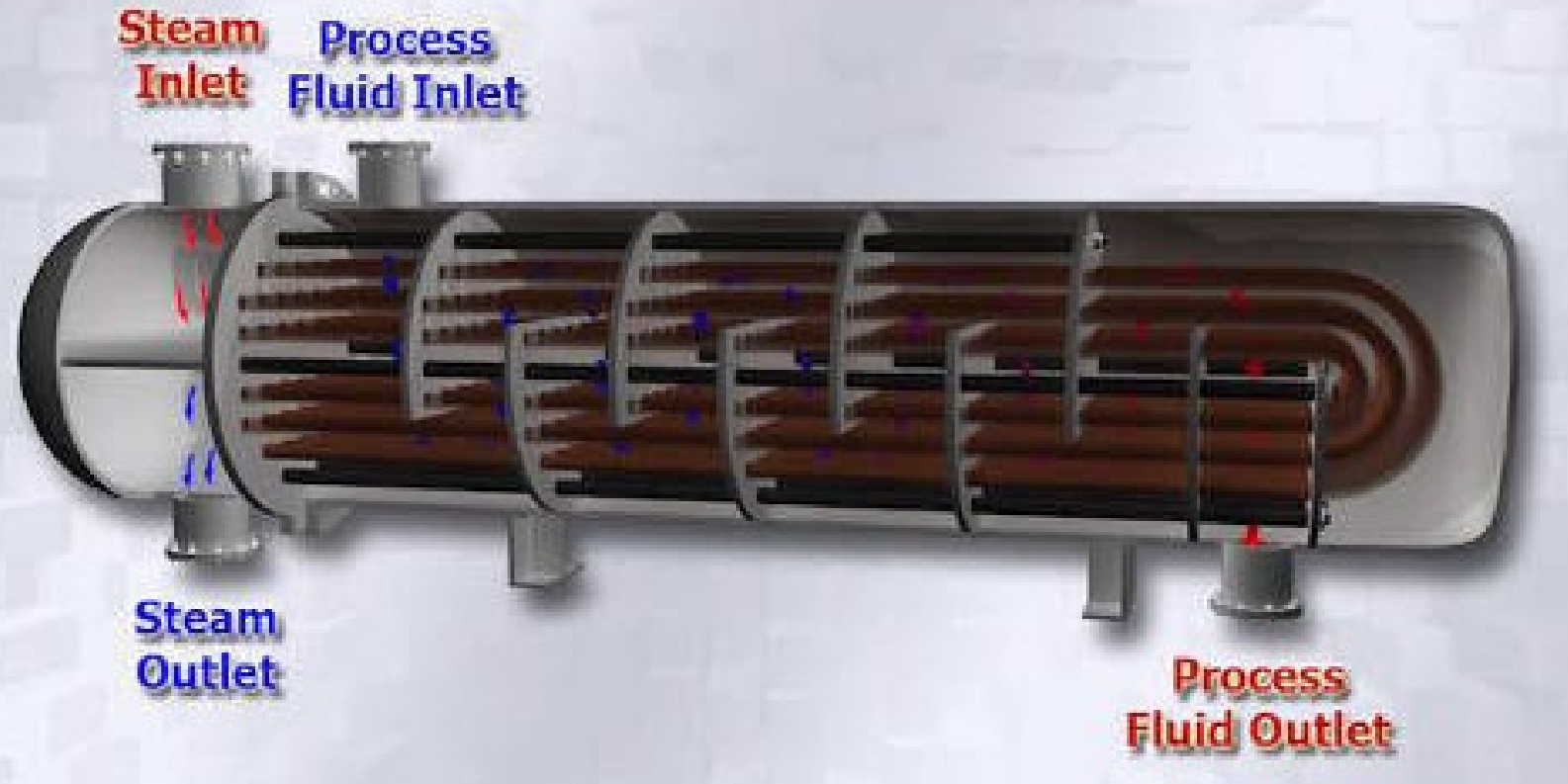




ME 300:
Engineering Thermodynamics

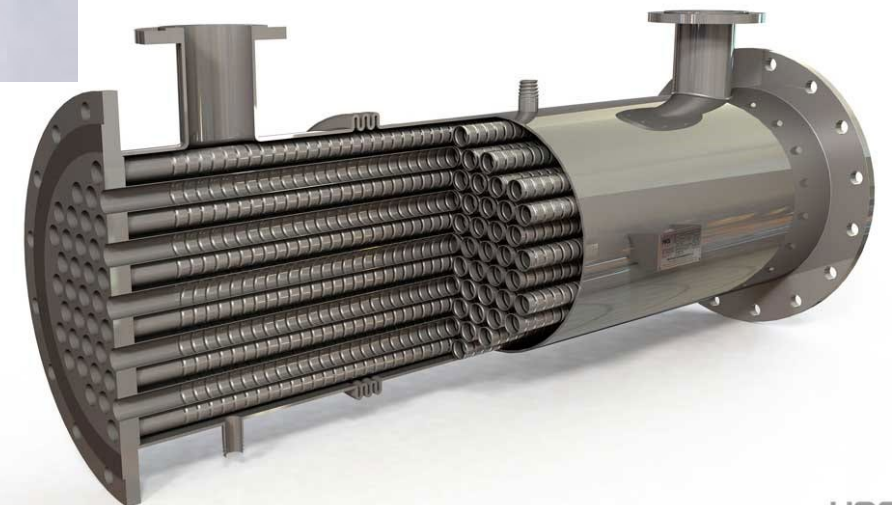
Heat Exchangers

Heat Exchangers



<https://www.controleng.com/articles/applying-heat-exchanger-control-strategies/>

<https://www.hrs-heatexchangers.com/us/image-library/>



Heat Exchangers

Assumptions: $\dot{W} = 0$

$$\Delta KE = 0$$

$$\Delta PE = 0$$

$\dot{Q}$ is not lost surroundings

$$\dot{Q}_{in, fluid 1} = - \dot{Q}_{out, fluid 2}$$

$$\dot{m} (\Delta h + \cancel{\Delta KE} + \cancel{\Delta PE}) = \dot{Q} - \cancel{\dot{W}}$$

$$\dot{m} \Delta h = \dot{Q}$$

$$\dot{m}_h (h_{h,2} - h_{h,1}) = - \dot{m}_c (h_{c,2} - h_{c,1})$$

Heat Exchangers